Information Retrieval

WS 2022 / **2023**

Lecture 10, Tuesday January 17th, 2023

(Classification, Naive Bayes)

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Overview of this lecture

- Organizational
  - Your experiences with ES9 — LSI, eigenvectors, etc.
  - Official course evaluation — more info next week

  Feel free to do the evaluation already now, but please:
  take your time and be **honest** and **fair** (and in a good mood)

- Contents
  - Classification — introduction and examples
  - Probability recap — two one-slide crash courses
  - Naïve Bayes — algorithm, example, implementation

  ES10: learn to predict the **genre** (Comedy, Docum, Horror, Romance, Sci-Fi) from a movie description using Naïve Bayes

Your experiences with ES9 1/3

- Summary / excerpts
  - Most of you found it very interesting and quite doable

    "Nice, definitely helped for understanding the lecture"

    "Very interesting lecture, matrices are more fun than logarithms"

    "The math sheet I enjoyed the most up to this point"

    "It was quite nice to improve the mental arithmetic"

    "My god, I hate pen and paper math … but it got better"

    "Took a long time … mixed feelings regarding calculations"

    "Fun doing Linear Algebra <u>and</u> seeing the practical uses"

    "A world without eigenvectors would be pretty eigenartig"

    "It is a shame that in modern capitalist societies everyone only thinks about their eigenvalues."

Your experiences with ES9 2/3

- Why are eigenvectors so fundamental
  - Symmetric matrices crop up naturally in many research areas (mathematics, computer science, physics, …)
  - The eigenvectors and eigenvalues characterize, what the matrix (viewed as a linear mapping) does to the space:

    It stretches it in some dimensions and squishes it in other dimensions (the factors are the eigenvalues)

    By leaving out the dimensions where it's stretched less (or squished), one gets a natural dimension reduction

    For example, consider the repeated application and normalization of a symmetric matrix A to a vector x:

$S = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$

$S^2 = \begin{pmatrix} \lambda_1^2 & 0 \\ 0 & \lambda_2^2 \end{pmatrix}$

$S^k = \begin{pmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{pmatrix}$

$A = U \cdot S \cdot U^T$ eigendecomposition, U orthonormal, S diagonal

$A^2 = U \cdot S \cdot \underbrace{U^T \cdot U}_{I} \cdot S \cdot U^T = U \cdot S^2 \cdot U^T$, $A^k = U \cdot S^k \cdot U^T$

Your experiences with ES9 3/3

- YouTube videos on eigenvector intuition
  - 3Blue1Brown has beautifully made videos on many math topics, which give nice glimpses of intuition, for example:

    https://www.3blue1brown.com/lessons/quick-eigen
  - But **beware**

    Learning from videos is very, very, very dangerous

    They can be very entertaining and it may appear like you understood everything (similar to our lectures maybe)

    But you only really understand by **doing it yourself**

    Without doing the work yourself, you very quickly forget (and never really understood it in the first place)

    A great movie scene ("A Serious Man") making the point

Classification 1/6

- Problem
  - Given **objects** and **classes**
  - Goal: given an object, predict to which class it belongs
  - To achieve that, we are given a **training set** of objects, each labeled with the class to which it belongs
  - From that we can (try to) learn which kind of objects belong to which class
  - Two examples on the next two slides

Classification 2/6

- Example 1 (natural language text)
  - **Training** set of documents, each labeled with its class

    In a small island off the American coast, the Whateleys live in an old mill where a mysterious bloody being … Horror

    A starship crew in the 23rd century goes to investigate the silence of a distant planet's colony … Sci-Fi

    Two hearing protection product sales reps have mixed fortunes in the exercise of their trades. They first have to … Comedy

  - **Prediction**

    Professor Ayres discovers a secret in an ancient stone and when he opens a crypt, he revives … **which class?**

Classification 3/6

- Example 2 (artificial documents)
  - Training set of documents, each labeled with its class

aba	A
baabaaa	A
bbaabbab	B
abbaa	A
abbb	B
bbbaab	B

the "words" are letters here
(a or b, spaces omitted)
and we have two classes

  - Prediction

abababa	which class?
baaaaaa	which class?

Classification 4/6

- Difference to (hard or soft) clustering
  - Hard clustering: assign each object to exactly one cluster

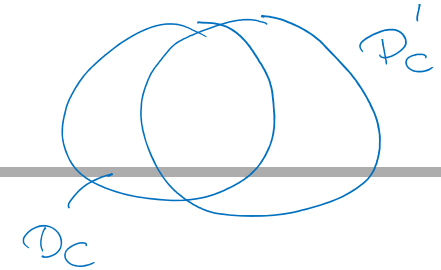
    A simple algorithm for hard clustering is k-means
  - Soft clustering: objects can belong to several clusters to different proportions

    For example, LSI can be seen as doing soft clustering (clusters correspond to the concepts from the m x k matrix)
  - **Difference 1:** The clusters have no pre-assigned "names"
  - **Difference 2:** There is no learning phase with given labels

    LSI does what is called **unsupervised** learning

    Naive Bayes does **supervised** learning

Classification 5/6



- Quality evaluation
  - Given a **test set** of labeled documents, and the predictions from a classification algorithm
  - For each class c, we consider the so-called ground truth and what the algorithm actually computes

    D_c = documents labeled c (in the test set)

    D'_c = documents classified as c (by the algorithm)
  - Then we can compute for each class

$$\frac{\frac{1}{P}+\frac{1}{R}}{2} = \frac{P+R}{2 \cdot P \cdot R} = \frac{1}{F_1}$$

„harmonic mean"

Precision $P := |D'_c \cap D_c| \,/\, |D'_c|$

Recall $R := |D'_c \cap D_c| \,/\, |D_c|$

F1-measure $F1 := 2 \cdot P \cdot R \,/\, (P + R)$ also called F1-score

Note that $P = R = F1 = 100\%$ if and only if $D_c = D'_c$

Classification 6/6

- Quality evaluation, difference to Lecture 2
  - Measures from Lecture 2: P@K, AP, nDCG, bpref, …
  - The setting there was (see L2, slide 29):

    The ground truth is a **set**

    The algorithms computed **ranked lists**
  - The setting in today's lecture is:

    The ground truth is a **set**

    Our algorithm computes a **set**

    In all of your future work, when you evaluate precision and recall, **always make explicit what these sets are**, otherwise it's hard to understand what exactly you evaluated

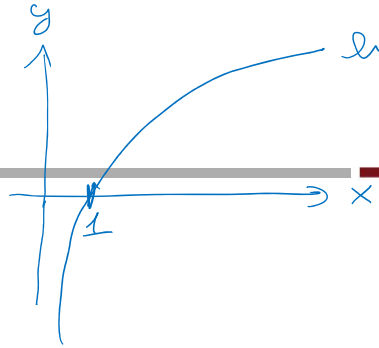
Probability recap 1/3

- Motivation
  - In this lecture, we will look at Naive Bayes, one of the simplest (and still widely used) classification algorithms
  - Naive Bayes makes **probabilistic assumptions**
  - For that, two very basic concepts from probability theory need to be understood:

    Maximum Likelihood Estimation (MLE)

    Conditional probabilities and Bayes Theorem
  - The following two slides are to refresh your memory concerning both of these

Probability recap 2/3



- Maximum Likelihood Estimation (MLE)
  - Consider a sequence of independent coin flips, for example:

    HHTTTTTTHTTTTTHTTHTT (5 times H, 15 times T)
  - Which Pr(H) and Pr(T) are the most likely?
  - Looks like Pr(H) = ¼ and Pr(T) = ¾ … let's prove this

$p = Pr(H) \Rightarrow Pr(T) = 1-p$

$Pr(HHTT\ldots HTT) = p^5 \cdot (1-p)^{15} \quad (*)$

Find p such that $(*)$ is maximized

Equivalently: find p such that $f(p) = \ln(p^5 \cdot (1-p)^{15})$ is max

$f(p) = 5 \cdot \ln p + 15 \cdot \ln(1-p)$

$f'(p) = \frac{5}{p} - \frac{15}{1-p}$

find p where $f'(p) = 0$, $p \neq 0$, $p \neq 1$

$f'(p) = 0 \iff \frac{5}{p} = \frac{15}{1-p}$

$\iff 5(1-p) = 15p$

$\iff 5 = 20p$

$\iff p = \frac{1}{4}$

CHECK that $f''(\frac{1}{4}) < 0$

$\Rightarrow$ max. at $p = \frac{1}{4}$ ∎

$\lim_{p \to 0} f(p) = -\infty$

$\lim_{p \to 1} f(p) = -\infty$

same because ln is a strictly monotone function

Probability recap 3/3

$Pr(A \cap B) = Pr(A \mid B) \cdot Pr(B)$
$Pr(B \cap A) = Pr(B \mid A) \cdot Pr(A)$

- Conditional probabilities

For an intuitive understanding, assume that Ω is finite, and all x in Ω are equiprobable

  - Let A and B be events in a probability space Ω
  - Denote by Pr(A | B) the probability of A ∩ B in the space B

    (1) Pr(A | B) := Pr(A ∩ B) / Pr (B)

    (2) Pr(A | B) · Pr(B) = Pr (B | A) · Pr(A)

  - The latter is called **Bayes Theorem**, after Thomas Bayes, 1701 – 1760

Random experiment: roll a die once

$\Omega = \{1, 2, 3, 4, 5, 6\}$, $\forall x \in \Omega \; Pr(x) = \frac{1}{6}$

$A = \{2, 4, 6\} \subseteq \Omega$ „even number"

$B = \{1, 2, 3\} \subseteq \Omega$ „small number ≤ 3"

$Pr(A) = \frac{|A|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}$; $Pr(B) = \frac{|B|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}$

$Pr(A \cap B) = \frac{|A \cap B|}{|\Omega|} = \frac{1}{6}$

$Pr(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{1}{3} = \frac{|A \cap B| \setminus |\Omega|}{|B| \setminus |\Omega|} = \frac{Pr(A \cap B)}{Pr(B)} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}$ ☒

Naive Bayes 1/13

- Probabilistic assumptions
  - Underlying probability distributions:

    A distribution p_c over the classes … where $\Sigma_c\, p_c = 1$

    For each c, a distr. p_{wc} over the words … where $\Sigma_w\, p_{wc} = 1$
  - Naïve Bayes assumes the following process for generating a document D with m words $W_1 \ldots W_m$ and class label C:

    Pick $C=c$ with prob. p_c , then pick each word $W_i=w$ with probability p_{wc} , **independent** of the other words
  - This is unrealistic in two respects (hence <u>Naïve</u> Bayes):

    1. Words in a document are not independent (for example, when "relativity" is present, "theory" is more likely)

    2. Word order is important, but is ignored by Naïve Bayes

Naive Bayes 2/13

- Learning phase
  - For a **training set** T of objects, let:

    T_c = the set of documents from class c

    n_{wc} = #occurrences of word w in documents from T_c

    n_c = #occurrences of all words in documents from T_c
  - We compute the following probabilities or likelihoods:

    $\mathbf{p_c := |T_c| \,/\, |T|}$ global likelihood of a class

    $\mathbf{p_{wc} := n_{wc} \,/\, n_c}$ likelihood of a word for a class

    The rationale behind these formulas is MLE … see slide 13

    Beware: n_{wc} and hence p_{wc} are often **zero** … see slide 21

Naive Bayes 3/13

- Learning phase, example
  - Consider Example 2 (artificial documents)

aba	A
baabaaa	A
bbaabbab	B
abbaa	A
abbb	B
bbbaab	B

$|T_A| = 3$, $|T_B| = 3$, $|T| = 6$

$P_A = \frac{|T_A|}{|T|} = \frac{1}{2}$, $P_B = \frac{|T_B|}{|T|} = \frac{1}{2}$

$n_{aA} = 10$, $n_{bA} = 5$, $n_A = 15$

$p_{aA} = \frac{n_{aA}}{n_A} = \frac{2}{3}$, $p_{bA} = \frac{n_{bA}}{n_A} = \frac{1}{3}$

$n_{aB} = 6$, $n_{bB} = 12$, $n_B = 18$

$p_{aB} = \frac{n_{aB}}{n_B} = \frac{1}{3}$, $p_{bB} = \frac{n_{bB}}{n_B} = \frac{2}{3}$

RESULT of the training:

$P_A = \frac{1}{2}$, $P_B = \frac{1}{2}$

$p_{aA} = \frac{2}{3}$, $p_{bA} = \frac{1}{3}$

$p_{aB} = \frac{1}{3}$, $p_{bB} = \frac{2}{3}$

Naive Bayes 4/13

$Pr(A \cap B) = Pr(A|B) \cdot Pr(B) = Pr(B|A) \cdot Pr(A)$

$Pr(A|B) = Pr(B|A) \cdot Pr(A) / Pr(B)$

- Prediction
  - For a given document **d** we want to compute the probability of each class **c**, given document **d**:

    Pr(C=c | D=d)

  - Using Bayes Theorem, we have:

    $Pr(C=c \mid D=d) = Pr(D=d \mid C=c) \cdot Pr(C=c) / Pr(D=d)$

  - Using our (naive) probabilistic assumptions, we have:

    $Pr(D=d \mid C=c) = Pr(W_1=w_1 \cap \ldots \cap W_m=w_m \mid C=c)$

    $= \Pi_{i=1,\ldots,m} Pr(W_i=w_i \mid C=c)$

    $= P_{w_1 c} \cdot P_{w_2 c} \cdot \ldots \cdot P_{w_m c}$

Naive Bayes 5/13

- Prediction … continued
  - We thus obtain that

    $\Pr(C=c \mid D=d)$

    $= \Pi_{i=1,\ldots,m} \Pr(W_i=w_i \mid C=c) \cdot \Pr(C=c) / \Pr(D=d)$

    $= \boldsymbol{\Pi_{i=1,\ldots,m}\, p_{w_i c} \cdot p_c} / \Pr(D=d)$
  - Note 1: for $\Pi_{i=1,\ldots,m}\, p_{w_i c}$ just take the p_{wc} for all words w in the document and multiply them (if a word w occurs multiple times, also take the factor p_{wc} multiple times)
  - Note 2: the $\Pr(D=d)$ is the same for all classes c

    We can hence compute the class c with the largest $\Pr(C=c \mid D=d)$ entirely from the learned p_{wc} and p_c

Naive Bayes 6/13

- Prediction, example
  - Consider Example 2 (artificial documents)

aba	A
baabaaa	A
bbaabbab	B
abbaa	A
abbb	B
bbbaab	B

  - Let us predict the class for aab … A or B ?

$$\frac{Pr(A \mid aab)}{Pr(B \mid aab)} = \frac{\frac{2}{3}\cdot\frac{2}{3}\cdot\frac{1}{3}\cdot\frac{1}{2}}{\frac{1}{3}\cdot\frac{1}{3}\cdot\frac{2}{3}\cdot\frac{1}{2}} = 2$$

$\Rightarrow$ PREDICT A

We learned during training:

$P_A = P_B = \frac{1}{2}$

$P_{aA} = \frac{2}{3}, \quad P_{bA} = \frac{1}{3}$

$P_{aB} = \frac{1}{3}, \quad P_{bB} = \frac{2}{3}$

Bayes Theorem

Independence assumption from „Naive Bayes"

$$Pr(A \mid aab) = Pr(aab \mid A) \cdot Pr(A) \,/\, Pr(aab)$$
$$= P_{aA} \cdot P_{aA} \cdot P_{bA} \cdot P_A \,/\, Pr(aab)$$
$$= \frac{2}{3}\cdot\frac{2}{3}\cdot\frac{1}{3}\cdot\frac{1}{2} \,/\, Pr(aab)$$

$$Pr(B \mid aab) = Pr(aab \mid B) \cdot Pr(B) \,/\, Pr(aab)$$
$$= P_{aB} \cdot P_{aB} \cdot P_{bB} \cdot P_B \,/\, Pr(aab)$$
$$= \frac{1}{3}\cdot\frac{1}{3}\cdot\frac{2}{3}\cdot\frac{1}{2} \,/\, Pr(aab)$$

Naive Bayes 7/13

$$\sum_w (n_{wc} + \varepsilon) = \underbrace{\sum_w n_{wc}}_{n_c} + \underbrace{\sum_w \varepsilon}_{\varepsilon \cdot \#voc.}$$

- Smoothing
  - Problem: $\Pr(C=c \mid D=d) = 0$ if only one single $p_{wc} = 0$

    This happens rather easily, namely when d contains a word that did not occur in the training set for class c
  - Therefore, during training we actually compute

    $p_{wc} := (n_{wc} + \boldsymbol{\varepsilon}) / (n_c + \boldsymbol{\varepsilon} \cdot \textbf{\#vocabulary})$

    This is like adding every word ε times for every class

    Note that the change in the denominator is necessary, so that the p_{wc} still sum to 1 for all w and a fixed class c

    For ES10, take ε = 1/10 … a larger ε would add too much noise for the short documents from our datasets

Naive Bayes 8/13

- Smoothing … continued
  - Since $p_{wc} = 0$ is a problem, what about $\mathbf{p_c = 0}$ for a class c?

    Then $Pr(C=c \mid D=d) = \text{something} \cdot p_c = 0$ for **any** document d, and hence class c would never be predicted
  - When does $p_c = 0$ happen?

    It happens if and only if $|T_c| = 0$, that is, when there is no document from class c in the training set
  - But if we did not see any document from a particular class c during training, we can learn nothing for that class, and we cannot meaningfully predict it
  - So $Pr(C=c \mid D=d) = 0$ is actually meaningful in that case and we do not need any additional smoothing to deal with that

Naive Bayes 9/13

11-bit mantissa
1 bit for the sign
+ subnormal representation

- Numerical stability
  - Problem: a product of many small probabilities quickly becomes zero due to limited precision on the computer

    For example, the smallest positive number that can be represented with an 8-byte double is $2^{-1074} \approx 5 \cdot 10^{-324}$

    So e.g. multiplying 54 probabilities $< 10^{-6}$ is already zero
  - Therefore, compute the **log**-probabilities … then products of probabilities translate into sums of log-probabilities

    Log-probabilities also give you the most likely class, because log is a strictly monotone function

    Don't take **exp** in the end, or you have the same problem again … for example, in Python3: math.exp(-746) = 0

Naive Bayes 10/13

- Some possible refinements
  - Instead of words, we could take any other quantifiable aspect of a document as so-called "feature"

    For example, also consider all (two-word) phrases
  - Don't use non-predictive words like "and" as features

    For example, omit the most frequent words or omit all words from a predefined list of so-called "stopwords"
  - In training, replace the word frequencies $\mathbf{n_{wc}}$ by $\mathbf{tf.idf_{wc}}$

    And correspondingly, replace $\mathbf{n_c}$ by $\mathbf{\Sigma_w\ tf.idf_{wc}}$
  - For ES10, these are all optional … however, you must use stopwords.txt to filter which words to show in the end for each class

Naive Bayes 11/13

- Linear algebra (LA)
  - Assume the documents are given as a term-document matrix, like we have seen it in Lecture 8 and 9

    For ES10, we provide you with the code to construct the document-term matrix with simple tf entries
  - Then all the necessary computations can be done very elegantly and efficiently using matrix operations

    Whenever you have to compute a large number of (weighted) sums in a uniform manner, this calls for LA

    On the next slide, see how it can be done for prediction

    Try to find out yourself how to do it for the training (a bit more complicated than prediction)

Naive Bayes 12/13

m is now the size of the vocabulary

NOTE: if $tf_i = 0$, then $p_{ic}^{tf_i} = 1$

- Linear algebra (LA)
  - For example, let's try to cast the computation of the probability for a class c for a document d into LA:

    $Pr(C=c|D=d) = \Pi_{i=1,\ldots,m}\, p_{ic}^{tf_i} \cdot p_c \,/\, Pr(D=d)$

    where tf_i is the number of occurrences of w_i in D
  - Take the logarithm and omit $Pr(D=d)$ (it's the same for all c)

    $\Sigma_{i=1,\ldots,|V|}\, tf_i \cdot \ln p_{ic} + \ln p_c$
  - This is just the dot product of these two vectors from $\mathbf{R}^{m+1}$

    $\mathbf{d} = (tf_1, \ldots, tf_m, 1)$ and $\mathbf{p_c} = (\ln p_{1c}, \ldots, \ln p_{mc}, \ln p_c)$
  - You can compute the dot products for all classes c simultaneously with a <u>single</u> matrix-vector product

Naive Bayes 13/13

- Useful numpy functionality for ES10
  - All of the following can be done with a matrix in one line:

    Multiply with a vector or transpose

    Set values for given lists of row and column indices

    Component-wise addition or multiplication of another matrix

    Apply operation to each entry, for example + or * or log

    Compute all row or column sums

    Compute largest or smallest value in each row or column

    Create from multiple copies of a vector

    See the numpy cheat sheet on our Wiki … when you wonder whether an operation exists, google it, it most certainly does

References

- Further reading
  - Textbook Chapter 13: Text classification & Naive Bayes

    http://nlp.stanford.edu/IR-book/pdf/13bayes.pdf
  - Advanced material on the whole subject of learning

    Elements of Statistical Learning, Springer 2009
- Wikipedia
  - http://en.wikipedia.org/wiki/Naive_Bayes_classifier
  - http://en.wikipedia.org/wiki/Bayes'_theorem
  - http://en.wikipedia.org/wiki/Maximum_likelihood
  - https://en.wikipedia.org/wiki/Double-precision